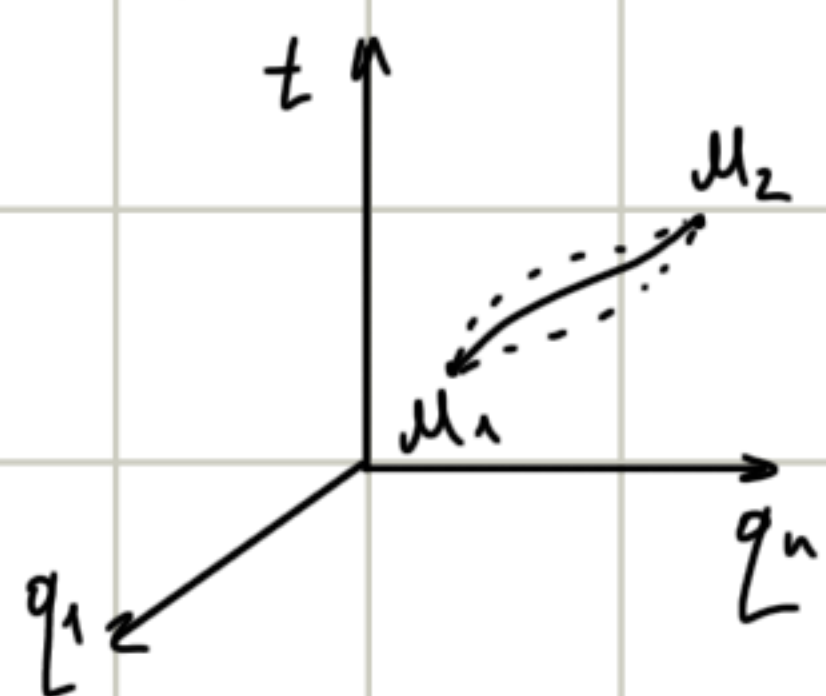


Принцип наименьшего действия



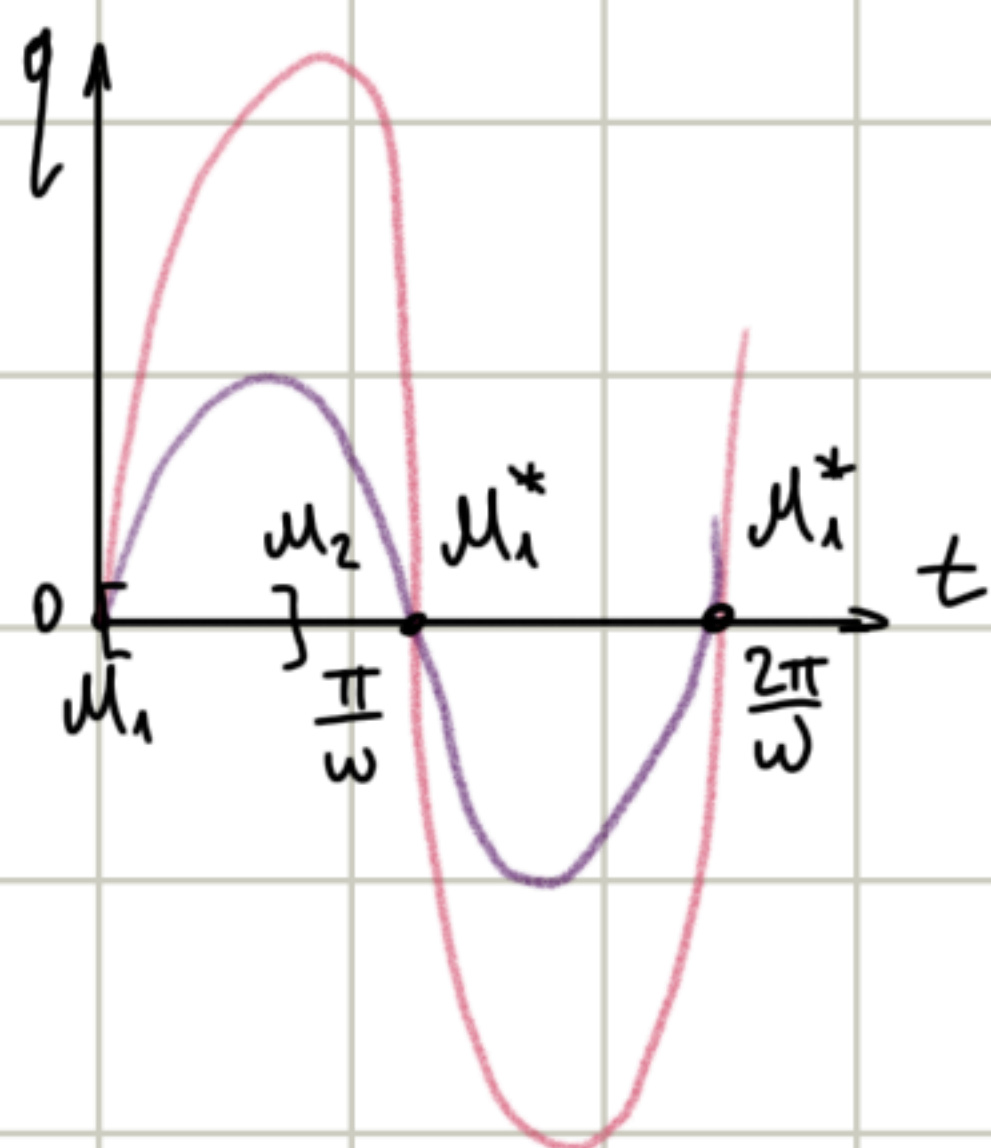
$$W = \int_{t_1}^{t_2} L dt$$

$$\delta W = 0 \Leftrightarrow \text{прямой путь}$$

th Лагранжа

Если на m_1, m_2 нет сопряж. кин. фокусов для m_1 , то W на прямом пути принимает min значение по ср. с окрестностями, если же кин. фокус есть, то W - не обяз. min

Кин. фокус m_1^* к m_1 - точки, в кот. нарушается единств. пр. пути



$$\ddot{q} + \omega^2 q = 0$$

Замена переменных в ур-х Лагранжа

$$\begin{cases} q_i^* = \varphi_i^*(\bar{q}, t) \\ t^* = \psi^*(\bar{q}, t) \end{cases}$$

замена
обратима

$$\begin{cases} q_i = \varphi_i(q^*, t^*) \\ t = \psi(q^*, t^*) \end{cases}$$

$$L(q, \dot{q}, t) \rightarrow q(t) - \text{прямой путь}$$

↓ замена пер.

$$\int_{\text{min}} L^*(q^*, \frac{dq^*}{dt^*}, t^*) \leftarrow q^*(t^*) - \text{прямой путь}$$

$$W = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$$

$$W^* = \int_{\varphi_1}^{\varphi_2} L(\varphi, \frac{d\varphi}{d\psi}, \psi) d\psi = \int_{t_1^*}^{t_2^*} \underbrace{L(\varphi, \frac{d\varphi}{d\psi}, \psi)}_{L^*} \frac{d\psi}{dt^*} dt^*$$

$$L^*(q^*, \frac{dq^*}{dt^*}, t^*) = L(\varphi(q^*, t^*), \frac{d\varphi}{d\psi}(q^*, t^*), \psi(q^*, t^*)) \cdot \frac{d\psi}{dt^*}$$

$$\frac{d\psi}{dt^*} = \frac{d\psi(q^*, t^*)}{dt^*} = \sum_{i=1}^n \frac{\partial \psi}{\partial q_i^*} \frac{dq_i^*}{dt^*} + \frac{\partial \psi}{\partial t^*}$$

$$\frac{d\varphi_i}{d\psi} = \frac{d\varphi_i/dt^*}{d\psi/dt^*} = \frac{\sum_{i=1}^n \frac{\partial \varphi_i}{\partial q_i^*} \frac{dq_i^*}{dt^*} + \frac{\partial \varphi_i}{\partial t^*}}{\sum_{i=1}^n \frac{\partial \psi}{\partial q_i^*} \frac{dq_i^*}{dt^*} + \frac{\partial \psi}{\partial t^*}}$$

Инвариантность лагранжиана

$$L^*(q^*, \frac{dq^*}{dt^*}, t^*) = L(q^*, \frac{dq^*}{dt^*}, t^*)$$

th Лемма Нётер

преобразование:
$$\begin{cases} q_i^* = \psi_i^*(\bar{q}, t, \alpha) \\ t^* = \psi^*(\bar{q}, t, \alpha) \end{cases} \quad (*)$$

1) обратимо

2) переход при $\alpha=0$
$$\begin{cases} q_i^* = \psi_i^*(\bar{q}, t, 0) = q_i \\ t^* = \psi^*(\bar{q}, t, 0) = t \end{cases}$$

th Лемма Нётер

∃ сис. мат. м. гл. в попереч. поле и у сис. есть лагранжиан

$L(\bar{q}, \dot{\bar{q}}, t)$. ∃ преобр. (*) со св-ми 1) и 2): L -инвариантен отн.

преобр., тогда ∃ ПУ: $\Phi(q, p, t) = \sum p_i \frac{\partial \psi_i^*}{\partial \alpha} \Big|_{\alpha=0} - H \frac{\partial \psi^*}{\partial \alpha} \Big|_{\alpha=0}$

Δ. $L^*(q^*, \frac{dq^*}{dt^*}, t^*) = L(\psi, \frac{d\psi}{dt^*}, \psi) \frac{d\psi}{dt^*}$ замена

$$L^*(q^*, \frac{dq^*}{dt^*}, t^*) = L(q^*, \frac{dq^*}{dt^*}, t^*)$$

$$L(\psi, \frac{d\psi}{dt^*}, \psi) \frac{d\psi}{dt^*} = L(q^*, \frac{dq^*}{dt^*}, t^*)$$

$$L(q, \dot{q}, t) = L(\psi^*, \frac{d\psi^*}{dt^*}, \psi^*) \frac{d\psi^*}{dt^*}$$

$\frac{\partial L}{\partial \alpha} = 0$

$$\left(\sum_{i=1}^n \frac{\partial L}{\partial q_i} \frac{\partial \psi_i^*}{\partial \alpha} + \frac{\partial L}{\partial \dot{q}_i} \frac{\partial}{\partial \alpha} \left(\frac{d\psi_i^*}{dt^*} \right) + \frac{\partial L}{\partial t} \frac{\partial \psi^*}{\partial \alpha} \right) \frac{d\psi^*}{dt^*} + L \cdot \frac{\partial}{\partial \alpha} \frac{d\psi^*}{dt^*} = 0, \quad \Big|_{\alpha=0}$$

$$\eta_i = \frac{\partial \psi_i^*}{\partial \alpha} \Big|_{\alpha=0}$$

$$\xi = \frac{\partial \psi^*}{\partial \alpha} \Big|_{\alpha=0}$$

$$\frac{d\psi^*}{dt^*} = \sum \frac{\partial \psi^*}{\partial q_i} \dot{q}_i + \frac{\partial \psi^*}{\partial t} \rightarrow \frac{d\psi^*}{dt^*} = 1 \quad (2)$$

$$\frac{\partial \psi^*}{\partial q_i} \Big|_{\alpha=0} = \lim_{\Delta q \rightarrow 0} \frac{\psi^*(q + \Delta q, t, \alpha) - \psi^*(q, t, \alpha)}{\Delta q} \Big|_{\alpha=0} = \lim_{\Delta q \rightarrow 0} \frac{\psi^*(q + \Delta q, t, 0) - \psi^*(q, t, 0)}{\Delta q} = 0$$

$$\frac{\partial \psi^*}{\partial t} \Big|_{\alpha=0} = \lim_{\Delta t \rightarrow 0} \frac{\psi^*(q, t + \Delta t, \alpha) - \psi^*(q, t, \alpha)}{\Delta t} \Big|_{\alpha=0} = 1$$

$$\frac{d\psi_i^*}{dt^*} \Big|_{\alpha=0} = \dot{q}_i \quad (1) \quad \text{аналогично}$$

$$\frac{d\psi_i^*}{dt^*} = \frac{d\psi_i^*}{d\psi^*} \frac{d\psi^*}{dt^*} \Big|_{\alpha=0}$$

$$\frac{d\psi^*}{d\psi^*} \Big|_{\alpha=0} = 1 \quad (3)$$

$$\frac{\partial}{\partial \alpha} \frac{d\psi^*}{dt^*} = \frac{\partial}{\partial \alpha} \left(\sum \frac{\partial \psi^*}{\partial q_i} \dot{q}_i + \frac{\partial \psi^*}{\partial t} \right) = \sum \frac{\partial^2 \psi^*}{\partial q_i \partial \alpha} \dot{q}_i + \frac{\partial^2 \psi^*}{\partial t \partial \alpha} = \frac{d}{dt} \frac{\partial \psi^*}{\partial \alpha}$$

$$\frac{\partial}{\partial \alpha} \left(\frac{d\varphi_i^*}{dt} \right) = \frac{\partial}{\partial \alpha} \left(\frac{d\varphi_i^*}{d\psi^*} \frac{d\psi^*}{dt} \right) = \frac{\partial}{\partial \alpha} \left(\frac{d\varphi_i^*}{d\psi^*} \right) \frac{d\psi^*}{dt} + \frac{d\varphi_i^*}{d\psi^*} \frac{\partial}{\partial \alpha} \frac{d\psi^*}{dt}$$

$$\Big|_{\alpha=0}$$

$$\frac{\partial}{\partial \alpha} \frac{d\varphi_i^*}{d\psi^*} \Big|_{\alpha=0} + \dot{q}_i \frac{d}{dt} \frac{\partial \psi^*}{\partial \alpha} \Big|_{\alpha=0} = \frac{d}{dt} \left(\frac{\partial \varphi_i^*}{\partial \alpha} \right) \Big|_{\alpha=0}$$

$$\frac{\partial}{\partial \alpha} \left(\frac{d\varphi_i^*}{d\psi^*} \right) \Big|_{\alpha=0} = \dot{q}_i - \dot{q}_i \dot{\xi} \quad (4)$$

$$\sum_{i=1}^n \frac{\partial L}{\partial \dot{q}_i} \cdot \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} (\dot{q}_i - \dot{q}_i \dot{\xi}) + \frac{\partial L}{\partial t} \xi + L \dot{\xi} = 0$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0$$

$$\frac{d}{dt} \left(\sum_{i=1}^n \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right) + \frac{d}{dt} \left[\left(L - \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right) \xi \right] = 0$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

$$H = \sum p_i \dot{q}_i - L$$

$$\Phi(q, p, t) = \sum p_i \frac{\partial \varphi_i^*}{\partial \alpha} \Big|_{\alpha=0} - H \frac{\partial \psi^*}{\partial \alpha} \Big|_{\alpha=0} \quad \blacktriangle$$

Законы сохранения

1) Обобщ. канон. мет. $\frac{\partial H}{\partial t} = 0 = \frac{\partial L}{\partial t}$

$$q_i^* = q_i$$

$$t^* = t + \alpha$$

$$L^* = L \cdot \frac{d\psi}{dt^*} \quad \text{--- 1}$$

$$L = L(q, \dot{q}) \quad \text{--- } t \text{ мет}$$

$$\Phi(q, p, t) = \sum p_i \cdot \frac{\partial \varphi_i^*}{\partial \alpha} \Big|_{\alpha=0} - H \frac{\partial \psi^*}{\partial \alpha} \Big|_{\alpha=0} = -H$$

$$\varphi_i^* = q_i$$

$$\psi^* = t + \alpha \quad \frac{\partial \psi^*}{\partial \alpha} = 1$$

2) q_1 - циклич. коорд.

$$\frac{\partial H}{\partial q_1} = 0 = \frac{\partial L}{\partial q_1}$$

$$q_1^* = q_1 + \alpha$$

$$q_j^* = q_j \quad j = \overline{2, n}$$

$$t^* = t$$

$$\varphi_1^* = q_1 + \alpha, \quad \varphi_j^* = q_j \quad j = \overline{2, n}, \quad \psi^* = t$$

$$\Phi(q, p, t) = p_1$$

Законы сохр. в замк. системах

$$\Pi(\bar{r}, t)$$

1) x_i, y_i, z_i

$$x_i^* = x_i + \alpha$$

$$\begin{aligned} y_i^* &= y_i \\ z_i^* &= z_i \\ t^* &= t \end{aligned}$$

$$L = \sum_{i=1}^n \frac{m_i}{2} (\dot{x}_i^2 + \dot{y}_i^2 + \dot{z}_i^2) - \Pi(\vec{r}, t)$$

$$P_{x_i} = \frac{\partial L}{\partial \dot{x}_i} = m_i \dot{x}_i$$

$$Q = \sum_i \frac{\partial L}{\partial \dot{x}_i} \cdot 1 = \sum m_i \dot{x}_i = Q_x$$

Аналогично для Q_y и Q_z

$$2) x_i^* = x_i \cos \alpha + y_i \sin \alpha$$

$$y_i^* = -x_i \sin \alpha + y_i \cos \alpha$$

$$z_i^* = z_i, \quad t^* = t$$

$$\left. \frac{\partial x_i^*}{\partial \alpha} \right|_{\alpha=0} = -x_i \sin \alpha \Big|_{\alpha=0} + y_i \cos \alpha \Big|_{\alpha=0} = y_i$$

$$\left. \frac{\partial y_i^*}{\partial \alpha} \right|_{\alpha=0} = -x_i \cos \alpha \Big|_{\alpha=0} - y_i \sin \alpha \Big|_{\alpha=0} = -x_i$$

$$Q = \sum P_{x_i} \cdot \left. \frac{\partial x_i^*}{\partial \alpha} \right|_{\alpha=0} + \sum P_{y_i} \cdot \left. \frac{\partial y_i^*}{\partial \alpha} \right|_{\alpha=0} = \sum_i m_i \dot{x}_i y_i - \sum_i m_i \dot{y}_i x_i = -k_z$$

мом. мом.

$$\begin{bmatrix} i & j & k \\ x_i & y_i & z_i \\ m_i \dot{x}_i & m_i \dot{y}_i & m_i \dot{z}_i \end{bmatrix}$$